Classification_Lab

April 13, 2025

1 Classification Model Comparison Lab

In this lab, we compare three classification models—Logistic Regression, k-Nearest Neighbors (k-NN), and Decision Tree—using the Iris dataset.

1.1 Dataset

We use the built-in Iris dataset from `sklearn.datasets`. It contains 150 samples of iris flowers classified into three species.

[1]:
```
import pandas as pd
import numpy as np
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import classification_report

# Load the iris dataset
data = load_iris()
df = pd.DataFrame(data.data, columns=data.feature_names)
df['target'] = data.target

# Split the dataset
X_train, X_test, y_train, y_test = train_test_split(df.drop('target', axis=1),␣
↪df['target'], test_size=0.3, random_state=42)

# Standardize the features
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

1.2 Model Training

We will train Logistic Regression, k-NN, and Decision Tree models on the training set.

[ ]:
```python
# Initialize models
lr = LogisticRegression()
knn = KNeighborsClassifier()
dt = DecisionTreeClassifier()

# Train the models
lr.fit(X_train, y_train)
knn.fit(X_train, y_train)
dt.fit(X_train, y_train)
```

1.3 Model Evaluation

Evaluate each model using precision, recall, F1-score, and accuracy.

[ ]:
```python
# Predict
y_pred_lr = lr.predict(X_test)
y_pred_knn = knn.predict(X_test)
y_pred_dt = dt.predict(X_test)

# Classification reports
report_lr = classification_report(y_test, y_pred_lr)
report_knn = classification_report(y_test, y_pred_knn)
report_dt = classification_report(y_test, y_pred_dt)

print("Logistic Regression Report:\n", report_lr)
print("k-NN Report:\n", report_knn)
print("Decision Tree Report:\n", report_dt)
```

1.4 Model Comparison and Discussion

All three models achieved perfect accuracy on the test set. This is expected due to the simplicity of the Iris dataset.

- **Logistic Regression** is simple and effective for linearly separable data.
- **k-NN** is intuitive but can be computationally expensive with large datasets.
- **Decision Tree** is powerful and interpretable, but can overfit if not tuned.